

Thesis_v0.9

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Abstract

Keywords

Introduction

Methods

Based on Okunogbe A, Nugent R, Spencer G, et al, this paper assesses the current and future impacts of overweight and obesity between 2020 and 2060. More specifically, a cost-of illness approach as described in D P Rice in “Estimating the cost of illness.”, splitting economic costs into direct and indirect costs is applied. This methodology is used to arrive the impact of obesity in monetary terms. It is useful for understanding the detrimental effects of obesity in terms of agenda-setting and policy prioritization.

The analysis includes estimates until 2050 and 2060, following the approach of Okunogbe A, Nugent R, Spencer G, et al. Estimations are done on a country-level basis of in total 182 countries. Data is sourced from publicly available databases, detailed information on the respective data and calculations is described in the corresponding section. A comprehensive overview of the data used can be seen in Table 1.1(insert table with used data)

By utilizing the cost-of-illness approach, the impacts of overweight and obesity are divided into direct and indirect cost. This thesis shall explain these costs and the performed calculations in a systematic manner, conveying a complex issue and extensive calculation in a comprehensive way. The paper is therefor divided into 3 sections:

1. Overweight and obesity prevalence
2. Direct costs
3. Indirect costs

The paper starts by inspecting the current situation of obesity prevalence and projecting this into 2060 by fitting autoregressive models to forecast future behavior based on past behavior data. We can then move on to direct and indirect costs, starting by closely inspecting direct costs. These include direct medical costs and direct non-medical costs. Indirect costs will be treated as the last part and consist of costs due to premature mortality and costs due to absenteeism and presenteeism.

Overweight and Obesity

Calculating any type of cost related to overweight and Obesity requires data on overweight and obesity prevalence. It is therefore a central object of this paper to predict future obesity prevalence in order to later use these predictions to arrive with estimates of the total economic costs related to overweight and obesity.

Data

Data from NCD Risk Collaboration group is sourced as a main basis for projecting future overweight and obesity prevalence. More specifically, this data includes country-level annual prevalence values for the years 1975 to 2016. To ensure a nuanced and detailed analysis, projections are made by age group and sex. Data from NCD Risk Collaboration thus includes one dataset for all males (referred to as men) and females (women) above 20 years of age, and one dataset for males (boys) and females (girls) below 20 years of age. Each group's prevalence is projected separately.

Furthermore, besides Age and Sex, Okunogbe A et al. identified *Demographic transition* and *Nutrition transition* as proxies for drivers of trends in overweight and obesity prevalence.

Demographic transition is presented by the proportion of the total population in aggregated age groups: 'under 20 years', '20-54 years', '54-74 years' and 'above 75 years'. We shall therefore denote it *demographic change* when modelling. Data on Population is sourced from United Nations Population Divisions, including past values from 1975-2016 and projections until 2060.

Nutrition transition is approximated by *urbanization*, measured as the percentage of population living in urban areas. As population tends to shift from rural to urban environments, there is also a shift in dietary habits and this dimension should be captured by incorporating urbanization into our model.

However, after importing the data from the United Population Division it appears that contrary to Okunogbe et al., data on urbanization is only available until 2050. Unfortunately, this limits the projections of the developed time series model as we need data until 2060 to project overweight and obesity prevalence until 2060 with the corresponding covariate.

The subsequent section will explain the modelling approach, creating an understanding of how the described factors are used in modelling and why urbanization sadly cannot be incorporated.

Modelling and Projections

Following the approach of Okunogbe A et al. in "Economic impacts of overweight and obesity: current and future estimates for eight countries" we estimated a multivariate autoregressive model. We intend to do this on a global scale, for 182 countries, extending the geographical aspect in the paper "Economic impacts of overweight and obesity: current and future estimates for eight countries".

The model employed retains the same structure as the one proposed by Okunogbe et al., taking the following form for *country i*, *year t* and *sex s*:

$$\Delta Y_{i,s,t} = \alpha + \sum_{h=1}^n \beta_h \Delta Y_{i,s,t-h} + X'_{i,s,t-1} \theta + \epsilon_{i,s,t}$$

where:

- $\Delta Y_{i,s,t}$ represents the differenced transformation of obesity prevalence with n degrees of lag.
- $X_{i,s,t}$ are external factors influencing obesity prevalence. That is, demographic changes and urbanization in our case. For demographic changes (Urbanization + Population) the age group 'under 20 years' was used as exogenous variable for boys and girls, and for male and female the remaining age groups (20-52 years, 54-74 years, 'above 75 years')
- α is the intercept.
- $\epsilon_{i,s,t}$ refers to the error term.

With the intention of producing reasonable values of obesity prevalence for future years and trying to reproduce the modelling approach by Okunogbe et al., we decided to stick with the same autoregressive order, degree of differencing, and external factors influencing the time series data. However, for external factors we could sadly only use Population change and had to dismiss the use of urbanization due to the lack of data.

The chosen lag of 2 degrees in the autoregressive component and first-order differencing of all variables ensures that our model accounts for past prevalence observations and captures the dynamics of demographic changes.

In the end, the model takes the following form:

$$\Delta Y_{i,s,t} = \alpha + \sum_{h=1}^2 \beta_h \Delta Y_{i,s,t-h} + X'_{i,s,t-1} \theta + \epsilon_{i,s,t}$$

(Okunogbe et al. 2021)

Furthermore, time series data (prevalence of overweight and obesity) is constraint to an upper bound of 100% using a logit transformation as expressed by the formula:

$$\log \left(\frac{\text{overweight_obesity}}{1 - \text{overweight_obesity}} \right)$$

(Okunogbe et al. 2021)

Values are then re-converted back before forecasting the prevalence in such a way, that the forecast in the end consist of ‘real’ numbers.

The intention is to use this for all 182 countries, but this is unfortunately not possible due to the nature of the data. While fitting the model was model was not a problem in most cases, there are some ‘outliers’ like Australia or Papua New Guinea for which stationary was not fulfilled completely. In these cases we implemented a second order of differencing to ensure stationarity. This adjustment not only facilitated successful model fitting and predictions but also mitigated errors like the one encountered, specifically *Error in stats::arima(x = x, order = order, seasonal = seasonal, xreg = xreg): non-stationary AR part from CSS.*

It is worth noting that the model parameters were set consistently with the specifications outlined in the reference paper that served as the basis for this analysis. Interestingly, the reference paper did not explicitly mention encountering any errors during their modeling process. While the model process is extended to 182 countries and only 8 of these countries are being incorporated by the author, the country Australia on one of these 8 and Australia is one of the cases where differencing of second order had to be applied.(Okunogbe et al. 2021)

Direct costs

Contribution

Question

Structure

Findings/Discussion

Conclusion

References

Appendix

Dataset Overweight and Obesity:

In order to work with the data, we make some adjustments to the dataset: underweight: $\text{bmi} < 18.5$ under-normal weight: $18.5 < \text{bmi} < 20$ normal weight: $20 < \text{bmi} < 25$ overweight: $25 < \text{bmi} < 30$ moderate obesity $30 < \text{bmi} < 35$ obesity: $\text{bmi} \geq 30$ severe obesity $\text{bmi} \geq 35$

oao for boys and girls:

oao prevalence projection: autoregression

exogenous variables

autoregression of 161 countries using $c(2,1,0)$ and using exception .notnull with $c(2,2,0)$ and $c(2,3,0)$

This is the model used to predict prevalence. I still have to experiment with the model parameters to find out the optimal combination. But for now I decided with to stick with these as the others did not work... So we fit an arimax(2,1,0) model and do this for men, women, girls and boys. We decided to leave out urbanization as a factor to enable us predicting until 2060. (data on urbanization only goes up to 2050)

Loop for men:

Loop for Women:

Loop for boys:

Loop for girls:

Regression of oafs

We now compute the average oao for 2020-2060 for each group. We can then use this to regress this average on the available average oaf of 2020-2050. This way, we can always plug in a number for obesity prevalence and predict OAFs from that:

requirements to fit oaf into projection dataframe:

regression for different groups:

Men:

Women:

Girls:

Boys:

weighted lm to predict oafs:

We now compute the weighted average prevalence of every year and country. This way, we can fit one general model and not one specific one for each group. We use this final model to predict oafs.

##creating a dataframe for all groups:

Computing the weighted oao:

Dataframe for average oao and average oao to use for regression

Direct medical costs

Costs of 2016:

costs till 2050/2060:

add oaf predictions to dataframe

add forecasts of health spending and predict direct costs:

#direct non-medical costs:

##travel cost: We calculated this by multiplying the average transport cost per trip by the population with overweight and obesity

We now add parameters for the number of outpatient and inpatient visits. “parameter sourced from a scoping review of peer-reviewed studies!” Which peer-reviewed studies? ” “For outpatient visits and hospitalizations (number and length), we use a global estimate due to the paucity of country-specific data” Which global estimate?

average inpatient/outpatient visits: (outpatient<-5.42, inpatient<-0.10)) https://www.researchgate.net/publication/329597721_Funding_and_services_needed_to_achieve_universal_health_coverage_applications_of_global_regional_and_national_estimates_of_utilisation_of_outpatient_visits_and_inpatient_admissions_from_1990_to_2016_a

HIC(spain):<https://pubmed.ncbi.nlm.nih.gov/28261758/> HIC(UK): <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6180713/> HIC(US):https://journals.lww.com/md-journal/fulltext/2016/01120/body_mass_index_trajectories_and_healthcare.54.aspx HIC(EU):[https://www.dovepress.com/the-humanistic-and-economic-burden-associated-with-increasing-body-mas-peer-reviewed-fulltext-article-DMSO](https://www.dovepress.com/the-humanistic-and-economic-burden-associated-with-increasing-body-mass-peer-reviewed-fulltext-article-DMSO) HIC(Germany/italy):<https://www.dovepress.com/obesity-in-germany-and-italy-prevalence-comorbidities-and-associations-peer-reviewed-fulltext-article-CEOR> LMIC(Brazil): <https://dmsjournal.biomedcentral.com/articles/10.1186/s13098-018-0322-9>

We need the income groups if we want to use this data on inpatient/outpatient visits:

addinpatient/outpatient:

calculate for 2019-2060:

Before we do that We include it into our predictions_oao dataframe. We can then calculate gasprice * 5km(average distance to facility)*0.06(1/km). We can then calculate inpatient and outpatient costs.

data clean up: Our dataframe is getting messy so lets do a clean up:

#indirect costs:

##premature mortality like described:

This is calculated as the number of years of potential life lost by individuals (by age group and sex) who died from OAO multiplied by the economic value of a life year.

life expectancy/mortality rate:

##abridged version:

Data on life expectancy:

oao related deaths (ord):

gdp per capita:

Calculate costs with Given Life expetancy:

test it with Australia (with given life expectancy) yll=life expectancy * deaths

try it with given life expectation from all countries (loop through countries):

formula in PDF:

Let's try it with the formula in like described (for australia:) The results are a bit better but still much more. . .

test it with australia (with formula used)

withour RR:

with RR:

##complete version:

oao related deaths (ord):

test it with australia male:

##premature mortality with calculated Ylls:

We use calculated YLLs from GDP 2019 for calculations. Due to missing data on RR we just projected the value form 2019 by multiplying it with the relative change of oao for the projected years. (?)

YLLs:

Let's compare the different methods:

men,women,boy,girls-wise:?

##reduced productivity due to additional absenteeism:

employment rates:

add abstenteeism:

“parameter sourced from a scoping review of peer-reviewed studies! <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4225558> https://journals.lww.com/joem/abstract/2014/01000/sick_leave_days_and_costs_associated_with.3.aspx

LIC: <https://pubmed.ncbi.nlm.nih.gov/29568332/> Specifically, for the normal BMI group the adjusted means for absenteeism was 6.21% and it was significantly lower than the overweight group (6.46; $p=0.02$), obesity class I (6.82; $p<0.001$), obesity class II (8.28; $p<0.001$), and obesity class III (10.03; $p<0.001$)

create weighted average: Among all respondents ($n=34,254$), a total of 46.4% were normal weight, 34.9% were overweight, and the remaining 18.7% were obese (12.9% were obese class I, 3.7% were obese class II, and 2.1% were obese class III)”

HIC: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0233512#pone-0233512-t001>

LIC:

HIC:

add these to the dataframe:

rename columns:

Wage predictions missing!

##reduced productivity due to additional presenteeism:

Okunogbe, Adeyemi, Rachel Nugent, Garrison Spencer, Johanna Ralston, and John Wilding. 2021. “Economic Impacts of Overweight and Obesity: Current and Future Estimates for Eight Countries.” *BMJ Global Health* 6 (10). <https://doi.org/10.1136/bmjgh-2021-006351>.